

## DTS 502

### Tank dosing station

ⒼⒷ Installation and operating instructions

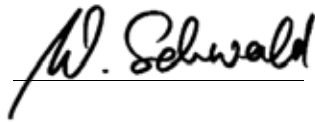


## Declaration of conformity

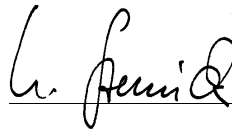
We **Grundfos Alldos** hereby declare under our sole responsibility that the products **DTS with electric agitator**, to which this declaration relates, comply with the following EC Directives for harmonisation with the legal requirements of the EC member states:

- Machinery (98/37/EC).
- Standards which have been used:  
EN ISO 12100, EN 60034-1 and EN 60204-1.
- Electromagnetic compatibility (89/336/EEC).
- Electrical equipment designed for use within certain voltage limits (73/23/EEC) [95].

Pfinztal, 10 December 2008



W. Schwald  
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### Warning

*These installation and operating instructions are also available online at [www.Grundfosalldos.com](http://www.Grundfosalldos.com).*



*Please read these installation and operating instructions before installation. Installation and operation must be undertaken in accordance with local regulations and the rules of technology.*

*Also observe the instructions provided for the components.*

## 1. General

### 1.1 Introduction

These installation and operating instructions, along with the relevant component instructions, contain all the information needed to commission and operate the DTS tank dosing station.

If you require further information or should problems arise which are not described in detail in this manual, please contact your nearest Grundfos Alldos branch.

### 1.2 Service documentation

If you have any questions, please contact your nearest Grundfos Alldos branch or service workshop.

## 2. Installation data

|      |
|------|
| Note |
|------|

*Please enter the following data after commissioning. It will help you and your Grundfos Alldos service partner to undertake settings on the installation later on.*

Owner:

Grundfos Alldos customer no.:

Order number:

Product number:

Serial no. of system:

Commissioning on:

Installation site:

Used for:

## 3. Installation sketch

## 4. General

### 4.1 Use

Tank dosing stations are used for storing and dosing liquid chemicals. They are standardised and can flexibly fulfil many dosing tasks. Through the use of high-quality materials, the stations can be universally used for many media without the material selection having to be adjusted.

Depending on the application requirements, the dosing pump can be selected from the DMS, DMI, DME series for up to 48 l/h and DDI series for up to 60 l/h and ordered separately.

### 4.2 Scope of delivery

DTS tank dosing stations always consist of a PE tank, suction unit with level switch and optional agitator with level switch. An injection unit, 10 m pressure line and multi-function valve can also be selected as optional extras.

DTS tank dosing stations can be configured and ordered with the assistance of the model key.

### 4.3 Guarantee

A guarantee claim in the sense of our general conditions of sale and delivery can only be recognised

- if the system is used as detailed in this manual.
- if the system is not dismantled or handled incorrectly.
- if repairs are only undertaken by authorised and qualified staff.
- if only original spare parts are used for repairs.

## 5. Safety notices

### 5.1 General

This operation manual contains general instructions that must be observed during installation, operation and maintenance. It must therefore be read by the installation engineer and the relevant qualified personnel/operators prior to installation and commissioning. It must be available at the installation location of the system at all times. It is not only the general safety notices described in this "Safety notices" section that must be observed, but all special safety notices that are provided in the other sections.

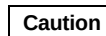
### 5.2 Identification of notices

If the safety notices or other instructions provided in this manual are not observed, this may result in personal injury or pump malfunction and damage. Safety notices and other information are identified by the following symbols:

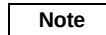


#### **Warning**

***Failure to observe these safety notices may result in injury.***



***Failure to observe these safety notices may result in malfunction and damage to property.***



***Information or instructions intended to simplify use and ensure safe operation.***

Information on the system itself, e.g. identification of fluid connections, must be clearly legible at all times.

### 5.3 Qualification and training of staff

The staff responsible for operation, maintenance, inspection and installation must be suitably qualified for this work. Areas of responsibility, responsibilities and supervision of personnel must be strictly controlled by the operator.

If staff do not have the necessary knowledge, they should be trained and instructed accordingly. If necessary, the training can be provided by the manufacturer/supplier at the request of the operator of the pump. The operator must ensure that staff understand the content of this manual.

### 5.4 Risks when safety notices are not observed

Non-observance of the safety notices may have dangerous consequences for staff, the environment and the system. Non-observance of the safety notices may lead to the loss of any claims for damages.

Non-observance of the safety notices may lead to the following risks:

- failure of important functions of the system
- failure of prescribed methods for maintenance
- personal injury caused by electrical, mechanical and chemical influences
- environmental damage due to media hazardous to health escaping.

## 5.5 Safety-conscious working

The safety notices described in this manual, applicable national regulations for health and safety and accident prevention and all internal working, operating and safety regulations for the operator must be observed.

## 5.6 Safety notices for the operator/user

Dangerous hot or cold pump parts must be protected to prevent accidental contact.

Any protection against accidental contact used for moving parts must not be removed when the system is operated.

Escaping hazardous media (e.g. hot, toxic) must be diverted such that they do not represent a hazard to people's health or the environment. Statutory regulations must be observed.

Damage caused by electrical energy must be prevented (for more details, please refer to the relevant regulations, for example those of the VDE and the local power supply company).

## 5.7 Safety notices for maintenance, inspection and installation work

The operator must ensure that all maintenance, inspection and installation work is undertaken by authorised, qualified staff who have been appropriately trained through reading this manual.

All work on the system may only be undertaken with the system stationary. The procedures described in this manual for bringing the system to a stop must be followed.

System parts which contain media hazardous to health must be decontaminated.

All safety and protection equipment must be put back into operation as soon as work is complete.

Observe the points described in the initial commissioning section prior to recommissioning.

### Warning



**Ensure that the system is suitable for the dosing medium used.**

**If using chemicals, observe the safety regulations provided by the manufacturer of these chemicals.**

### Warning



**Repairs may only be undertaken by authorised and qualified staff.**

**Wear protective clothing (gloves and goggles) when working on the system, connections or lines.**

### Caution

**The resistance of the parts that come into contact with media depends on the medium, media temperature and operating pressure. Ensure that parts which come into contact with media during operating conditions are resistant to the dosing medium.**

## 5.8 Unauthorised modification and manufacture of spare parts

Modifications or changes to the system are only permitted following agreement with the manufacturer. Genuine spare parts and accessories approved by the manufacturer are safe to use. Using other parts may result in liability for any damage incurred.

## 5.9 Improper operating methods

The operating safety of the system supplied is only ensured if used as described in section 6. *Technical data*.

The specified limit values should on no account be exceeded.

## 6. Technical data

### 6.1 Designations

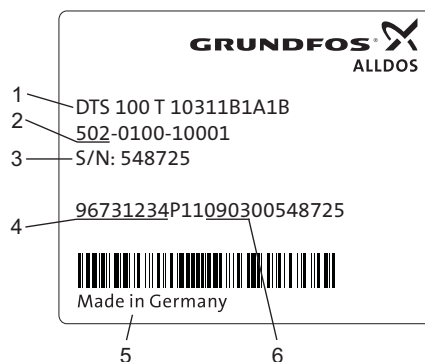


Fig. 1 DTS nameplate

| Item | Description            |
|------|------------------------|
| 1    | Type designation       |
| 2    | Model                  |
| 3    | Serial number          |
| 4    | Product number         |
| 5    | Country of manufacture |
| 6    | Code for year and week |

TM04 1227 5008

## 6.2 Type key

Example:

DTS 75T 0 1 2 2 A 9 D 2 B 1 D

### Tank dosing station

#### Tank

- 75** 75 litre tank
- 100** 100 litre tank
- 200** 200 litre tank
- 300** 300 litre tank
- 500** 500 litre tank
- 1000** 1000 litre tank
- T** transparent \*
- S** black \*

#### Catchment tray

- 0** without catchment tray
- 1** with catchment tray

#### Screw cover

- 0** Screw cover without lock

#### Handheld mixer / agitator

- 0** none
- 1** PVC handheld mixer
- 2** Electric agitator \*\* stainless steel, 220 - 240 V, 1-phase, 50/60 Hz (for 300 - 1000 l tank only 230/400 V, 50/60 Hz, 3-phase)
- 3** Electric agitator \*\* PP with sealing flange, 230/400 V, 1-phase, 50/60 Hz (for 300 - 1000 l tank only 230/400 V, 50/60 Hz, 3-phase)

**prepared for dosing pump (should be ordered separately)**

- 0** none
- 1** DMI 208, DDI 209, DMX (221) 4-10 to 50-10
- 2** DME 2 to 48, DMS, with adapter plate
- 3** DDI 60-10

#### Suction line for tank without electric agitator (flexible, with 2-stage empty message)

- A** none
- B** PVC, for DDI 0.4-10 to 5.5-10, DMI 0.3-10 to 6.0-8
- D** PVC, for DDI 13.8-4 to 20-3, DMI 9.0-6 to 18-3.5, DMX (221) 4-10 to 50-10
- F** PVC, for DDI 60-10
- H** PP, for DME / DMS 2 to 12
- J** PP, for DME 19 to 48

### Multi-function valve for fitting on dosing pump

- A** none
- B** for DDI 0.4-10 to 5.5-10, DMI 0.3-10 to 6.0-8
- C** for DDI 13.8-4 to 20-3, DMI 9.0-6 to 18-3.5, DMX (221) 4-10 to 50-10, DDI 60-10
- D** for DME / DMS 2 to 12
- E** for DDI 0.4-10... PVC-P3 to 5.5-10... PVC-P3

#### Filling unit

- 0** none
- 1** PVC / EPDM, with ball cock DN 15

#### Drain valve on tank

(can only be selected without catchment tray)

- A** none
- B** PVC / EPDM, DN 8

#### Injection unit (with ceramic ball and FKM O-ring)

- 0** none
- 1** PVC DN 8, for DDI and DMI, for DMX (221) 4-10 to 50-10, DDI 60-10
- 2** PP DN 8, for DME 19 to 48 and DMS 2 to 12

#### Pressure line, 10 m

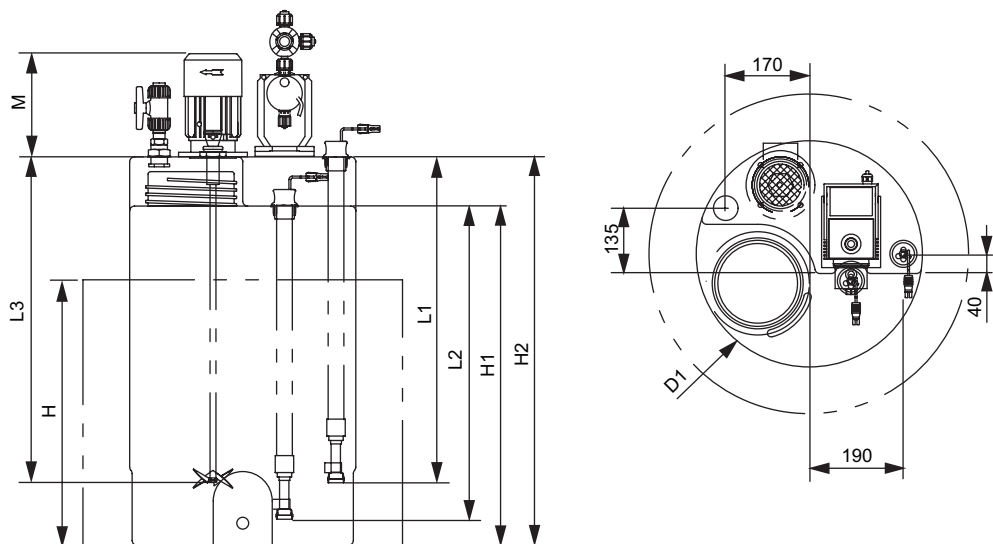
- A** none
- B** PE 4/6, for DME / DMS 2 to 12, DDI 0.4-10 to 5.5-10, DMI 0.3-10 to 6.0-8
- C** PVC 6/12, for DDI 13.8-4 to 20-3, DMI 9.0-6 to 18-3.5, DMX (221) 4-10 to 50-10
- D** PE 9/12, for DME 19 to 48

#### Suction line for tank with electric agitator (rigid, with 2-stage empty message)

- 0** none
- 1** PVC, for DDI 0.4-10 to 5.5-10, DMI 0.3-10 to 6.0-8
- 3** PVC, for DDI 13.8-4 to 20-3, DMI 9.0-6 to 18-3.5, DMX (221) 4-10 to 50-10
- 5** PVC, for DDI 60-10
- 7** PP, for DME / DMS 2 to 12
- 9** PP, for DME 19 to 48

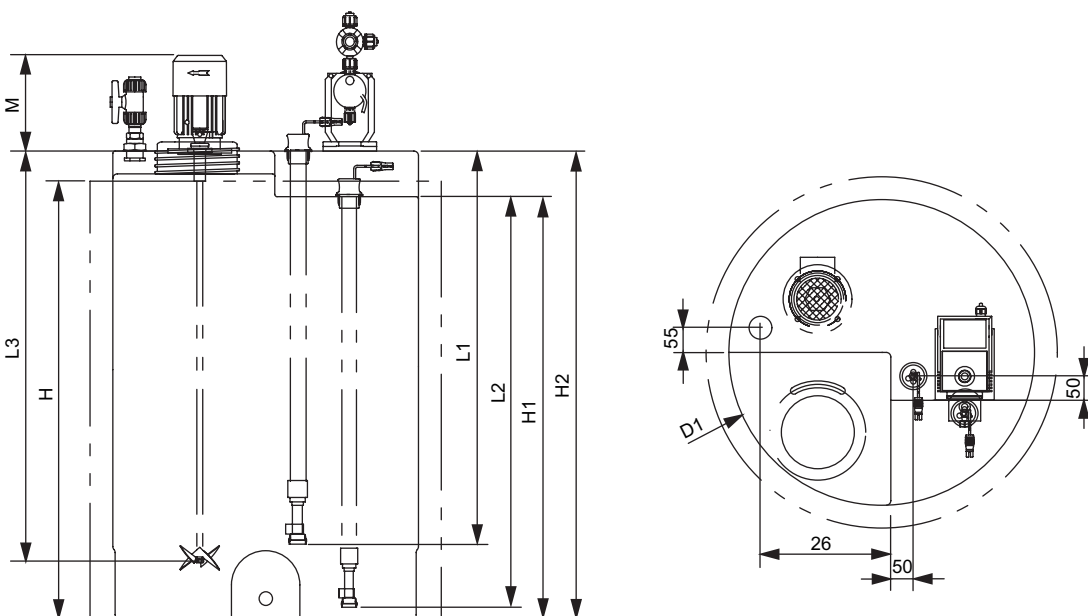
\* DTS tank dosing stations basically consist of a tank and at least one of the following components.

\*\* incl. level switch for deactivation in conjunction with separate control unit



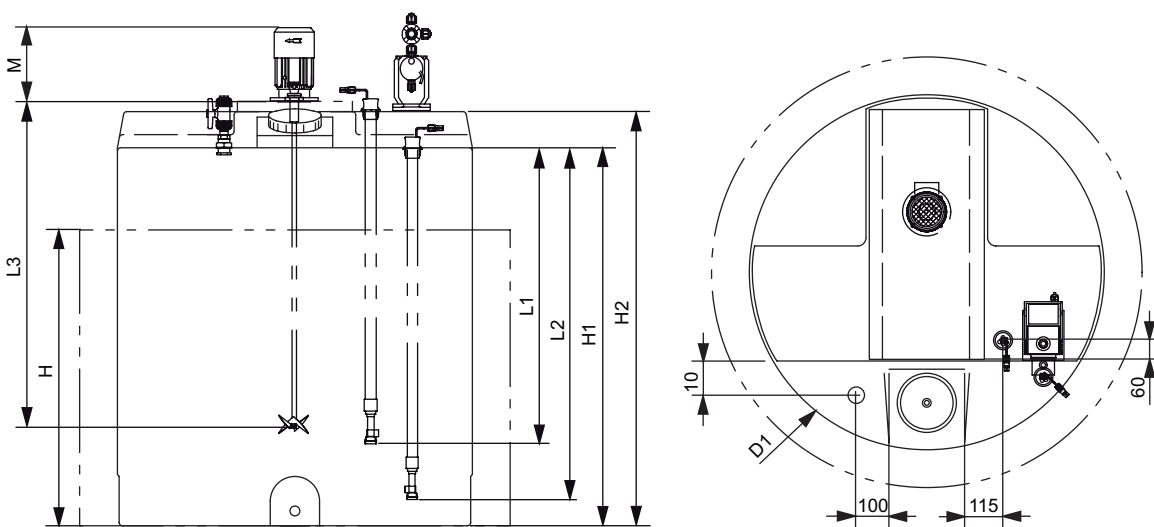
**Fig. 2** 75 / 100 l tank dosing station

TM04 1228 5008



**Fig. 3** 200 / 300 / 500 l tank dosing station

TM04 1229 5008



**Fig. 4** 1000 l tank dosing station

TM04 1230 5008



**Table of dimensions**

| Tank       |         |         |         | Level switch | Suction line | Agitator |        | Handheld mixer | Collection container |        |                     |
|------------|---------|---------|---------|--------------|--------------|----------|--------|----------------|----------------------|--------|---------------------|
| Volume [l] | D1 [mm] | H1 [mm] | H2 [mm] | L1 [mm]      | L2 [mm]      | L3 [mm]  | M [mm] | L [mm]         | H [mm]               | Ø [mm] | Useful capacity [l] |
| 75         | 460     | 530     | 630     | 430          | 500          | 500      | 210    | 480            | 540                  | 500    | 80                  |
| 100        | 460     | 690     | 790     | 600          | 660          | 650      | 210    | 600            | 700                  | 500    | 120                 |
| 200        | 600     | 780     | 880     | 660          | 750          | 750      | 210    | 750            | 730                  | 655    | 210                 |
| 300        | 670     | 930     | 1030    | 750          | 900          | 900      | 237    | 900            | 960                  | 770    | 400                 |
| 500        | 790     | 1080    | 1180    | 900          | 1030         | 1000     | 237    | —              | 980                  | 860    | 500                 |
| 1000       | 1080    | 1150    | 1260    | 900          | 1100         | 1000     | 234    | —              | 1080                 | 1150   | 1000                |

### 6.3 Materials in contact with media

- Dosing tank: PE
- Catchment tray: PE
- Handheld mixer: PVC
- Agitator
  - stainless steel (14.571)
  - PP
- Suction unit (suction line):
  - PVC/PE/ceramic, or
  - PP/PE/ceramic
- Pressure line:
  - PE, or
  - PVC
- Injection point:
  - PVC/FKM/ceramic
  - PP/FKM/ceramic
- Drain valve: PVC/EPDM
- Filling unit: PVC/EPDM
- Multi-function valve: PVDF/PTFE/FKM

### 6.4 Agitator motor performance data

| For tank | Motor type                                     | Degree of protection | Power [kW] | Speed [rpm] |
|----------|------------------------------------------------|----------------------|------------|-------------|
| 75 l     | Single-phase motor, 230 V, 50/60 Hz            | IP 54/B              | 0.09       | 1500        |
| 100 l    | Single-phase motor, 230 V, 50/60 Hz            | IP 54/B              | 0.09       | 1500        |
| 200 l    | Single-phase motor, 230 V, 50/60 Hz            | IP 54/B              | 0.09       | 1500        |
| 300 l    | Three-phase current motor, 230/400 V, 50/60 Hz | IP 65/F              | 0.25       | 1500        |
| 500 l    | Three-phase current motor, 230/400 V, 50/60 Hz | IP 65/F              | 0.25       | 1500        |
| 1000 l   | Three-phase current motor, 230/400 V, 50/60 Hz | IP 65/F              | 0.55       | 1500        |

For more details, refer to instructions provided for components.

7. System features

7.1 Purpose of DTS tank dosing station

- For dosing certain liquid dosing media in hydraulic systems.
- The dosing system is not intended for dosing gaseous or solid media.
- The dosing system must not be operated in conditions other than those described in the technical data.
- The dosing system may only be operated by technical staff



**Warning**  
*Any usage other than that described here is not intended. Grundfos Alldos accepts no liability for any damage resulting from incorrect use. The operator alone bears all risk in such cases.*

7.2 Dosing media

Thin (max. 200 mPas), non-explosive dosing media without abrasive or long fibre components. The dosing medium must not chemically attack the system materials.  
Max. permissible dosing medium temperature: 45 .

Note

*For chemical resistance, please refer to section 14.3 Assistant for selecting material combinations.*

7.3 Overview of device

7.3.1 Dosing system

- DTS tank dosing stations can comprise the following modules (selection depending on model key):
- Chemical-resistant tank made from UV-stabilised semi-transparent or black polyethylene, in 6 sizes between 75 and 1000 litres, sintered threaded bushes and/or adapter plate for installing a dosing pump, embossed litre scale, screw cover made from PE
  - catchment tray made from PE in various sizes for tanks of between 75 and 1000 litres
  - handheld mixer or electric agitator with level switch
  - flexible or rigid suction line made from PVC or PP with foot valve and 2-stage level switch as idling protection
  - injection unit\* made from PVC or PP with G 1/2" screw-in thread
  - 10 m pressure line\* made from PE or PVC
  - drain valve\*
  - filling unit
  - multi-function valve\*
  - dosing pump\* (order separately)

\* These components are prepared for subsequent installation and enclosed with the delivery packaged separately. The complete installation should be undertaken on site.

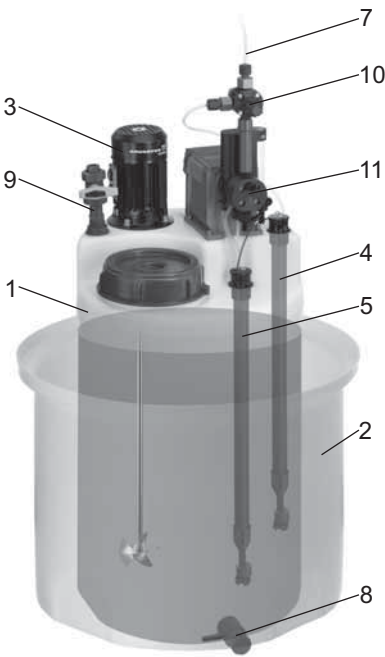


Fig. 5 Fitted DTS tank dosing station (example)

| Item | Description                |
|------|----------------------------|
| 1    | Tank                       |
| 2    | Catchment tray             |
| 3    | Agitator                   |
| 4    | Level monitoring           |
| 5    | Suction line               |
| 6    | Injection unit (not shown) |
| 7    | Pressure line              |
| 8    | Drain valve                |
| 9    | Filling unit               |
| 10   | Multi-function valve       |
| 11   | Dosing pump                |

## 8. Transport and storage

**Do not throw or drop the tank dosing station.**  
**Do not use the protective packaging as packaging for transport.**  
**Only transport the tank dosing station if the dosing pump is disassembled.**

### Caution

### 8.1 Unpacking

- Unpack the tank dosing station as soon as you receive it and check for visible transport damage.
- Retain the packaging for future storage or return, or dispose of the packaging in accordance with local regulations.

### 8.2 Storage

Permissible storage and transport temperature 0 °C to 50 °C.

## 9. Installation

**The dosing system may contain water from the check carried out in the factory. When dosing media which must not come into contact with water, we would recommend first using the pump to pump a different medium to remove water from the dosing head before installation.**

### Caution

### 9.1 Installation site

The dosing system should be installed in the catchment tray intended for it at a site suitable for the corresponding loads. The installation site should be horizontal, level and frost-free.

The dosing system must be easily accessible.

The system is mounted and wired virtually ready for connection in line with the details provided in the order.

Direct sunlight must be avoided since the dosing system's materials may be damaged by sunlight.

When installing the dosing system outdoors, roofing or similar protection should be provided to protect the dosing system from rain and other weathering.

**Before starting work please check whether all the technical conditions required at the installation site and on the system (see nameplate) are fulfilled.**

### Caution

## 9.2 Mounting and installing hydraulically

### Warning

**The dosing medium is pressurised and can be harmful. Note the maximum permissible pressure values.**

**When working with chemicals, the accident prevention regulations applicable at the installation site and the technical rules applicable to working with chemicals should be applied (e.g. wearing protective clothing).**



### Warning

**Before working on the dosing pump and system, mains cables must be disconnected and secured to prevent them being switched on again. Before switching the supply voltage back on, the dosing lines must be connected up such that any chemicals in the dosing system cannot spray out and put people at risk.**



### Warning

**When using the multi-function valve: Overflowing dosing medium must always be returned to a tank.**

- Media such as peracetic acid and hydrogen peroxide must be kept in a separate tank.
- Other media can be returned to the dosing tank.

**The overflow hose provided with the multi-function valve must be connected for this purpose and routed to the corresponding tank or the suction line.**



### Warning

**When changing chemicals you will need to check the materials used for chemical resistance on the dosing pump and the rest of the system. If there is a risk of a chemical reaction between the various media, the pump and system must be cleaned thoroughly before the new chemical is used.**



Depending on the scope of supply, the customer must mount or install the following modules.

### 9.2.1 Installation of dosing pump

**The retaining material for the dosing pump (screws, nuts, shims) is enclosed separately with tank dosing stations with the "prepared for dosing pump" option.**

### Note

- Mount the dosing pump and suitable retaining material directly on the tank or adapter plate.

### 9.2.2 Installation of multi-function valve

The multi-function valves are designed as 3-way-valves. They have an input, an output on the pressure side and an unpressurised overflow output.

- Fit the multi-function valve directly on the pressure valve of the dosing pump head.
- The optional extension adapter must be screwed between the valves on DMI and DDI model pumps in the version with Plus3 system.

Refer to separate operating instructions for multi-function valve.

### 9.2.3 Installing dosing lines

**Route hosing such that it is free from mechanical tension and bends.**

**- To ensure that the connections hold well, only use the clamp rings and hose connectors intended for the hose diameter in question.**

#### Caution

**- Only use genuine hoses with the prescribed hose dimensions and wall thicknesses.**  
**- Always observe the maximum permissible operating pressure for the dosing pump and line system.**

Connect suction line to suction valve and pressure line to pressure valve or multi-function valve on dosing pump.

1. Cut hose ends to length (straight cut).
2. Pull union nut and clamp ring over hose.
3. Slide hose end over connector to stop, widen if necessary and depending on the type of connection secure with a counterpiece or hose clip.
4. Fit gasket.
  - Ensure that the O-ring or flat gasket is positioned correctly in the counterpiece (pump valve/injection unit).
5. Use union nut to screw hose on valve.

Refer to separate operating instructions for the dosing pump supplied

### 9.2.4 Installation of injection unit

- Screw injection units vertically from above into the coupling thread (provided by the customer) of the process line.

Refer to separate operating instructions for injection unit.

### 9.2.5 Drain valve R 3/4"

- Fit the drain valve in the dosing tank's coupling thread.
  - Unscrew stopper
  - Drill open threaded hole
  - Screw in drain valve and seal with Teflon tape.

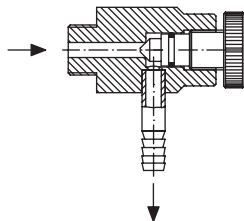


Fig. 6 Drain valve

### 9.3 Tightness test

1. Before filling the dosing tank check whether:
  - the suction line is connected, and
  - the optional drainage unit is fully closed.
2. Then fill the dosing tank with e.g. water and check for leaks.

#### Note

**For chemical resistance, please refer to section 14.3 Assistant for selecting material combinations.**

## 9.4 Electrical connection

#### Warning

**Electrical connections may only be established by trained staff.**

**Observe the local safety regulations.**

**Protect the cable connections and plugs against corrosion and humidity.**

**Before connecting the mains cables check whether the supply voltage stated on the nameplates of the dosing pump and agitator matches the local figures (permissible mains frequency deviation:  $\pm 5\%$ ). An incorrect mains voltage may destroy the components.**

#### Caution

- Fuse motor with a motor overload switch of the appropriate rating.

### 9.4.1 Electrical connections for dosing pump

Refer to separate operating instructions for the dosing pump supplied (should be ordered separately)

### 9.4.2 Electrical connections for agitator

#### Warning

**Before opening housing, disconnect agitator from power supply.**

1. Connect motor following circuit diagram in terminal box.
2. Fuse motor with a motor overload switch of the appropriate rating.
3. Check direction of rotation before commissioning (note arrow indicating direction of rotation).

### 9.4.3 Connection for level switch

If present, the suction unit and agitator are each fitted with a level switch.

1. Plug the connector on the suction line's level switch into the corresponding bush on the dosing pump.
2. The separate level monitoring can be used via an external control unit to deactivate the agitator when the tank is running empty.

### 9.4.4 Inputs / outputs

Refer to separate operating instructions for the dosing pump supplied.

Refer to separate operating instructions for suction line.

## 10. Commissioning

#### Warning

**With electric agitator only: never reach into the dosing tank when the electric agitator is running. The rotating propeller and stirring spindle may result in serious hand injuries.**

#### Caution

**Observe the dosing pump operating instructions.**

**With electric agitator only: before switching on the electric agitator, fill the dosing tank with dosing medium to at least 20 cm above the propeller.**

#### Caution

**If this is not done, turbulence may occur when stirring and the stirring spindle may be knocked.**

**The point at the top of suction unit where the suction line and level cable emerge must not be blocked or sealed.**

#### Caution

**It must be possible for air to enter here to compensate for pressure in the dosing tank.**

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## 11. Maintenance

The tank dosing station requires no maintenance. The dosing pump must be maintained regularly as stated in its instructions. Refer to separate operating instructions for the dosing pump supplied.

In the event of contamination or deposits, the agitator (mixer spindle and mixer) can be cleaned:

### 11.1 Cleaning the agitator



**Warning**

*The entire system should be switched off before any maintenance work.*

*Disconnect supply voltage before opening housing.*



**Warning**

*The agitator may only be removed and fitted by a qualified specialist.*

*In particular protect the mixer spindle and mixer from knocks, impact and twisting.*

*When removing or introducing into the tank, the mixer spindle must not be strained by the weight of the drive.*

**Caution**

*Do not hold or transport the agitator by the spindle or mixer.*

#### 11.1.1 Removal

1. Remove electrical connections and level monitoring if necessary.
2. Loosen the agitator's 4 retaining bolts.
3. Lift agitator vertically out of the dosing tank.
  - Do not twist mixer spindle or strain with the weight of the drive.

#### 11.1.2 Cleaning

1. Use appropriate cleaning agents to clean mixer spindle and mixer.

#### 11.1.3 Installation

1. Install agitator vertically on the dosing tank using 4 screws.
  - When guiding into tank do not twist mixer spindle or strain it with the weight of the drive.
2. After installation check that the mixer spindle and mixer are not touching any point of the tank.
3. Before commissioning re-establish electrical connections and level monitoring with level switch, check direction of rotation.

## 12. Service / spare parts / accessories

We would like to draw your attention to the fact that spare parts and accessories not supplied by ourselves have not been tested or approved by ourselves.

Under certain circumstances the installation and/or use of such products can therefore change for the worse or impair properties of the DTS tank dosing station resulting from its design.

There are no grounds whatsoever for liability or warranty on the part of Grundfos Alldos for damage resulting from the use of non-genuine spare parts or accessories.

Faults which cannot be remedied by the customer should only be remedied by the Grundfos Alldos service or authorised specialist firms.

Should a fault arise please provide an accurate description of the problem to enable our service technician to prepare and bring the right spare parts.

Please refer to the nameplate for the system's technical data.

## 13. Disposal

This product and all its associated parts must be disposed of in an environmentally friendly manner.

1. Suitable private waste disposal operations can be used here.
2. If no such operations exist or if they refuse to process the waste, the product can be sent to the closest Grundfos or Grundfos Alldos branch or service centre.
3. All traces of chemicals must be removed from the product before disposal.

## 14. Annex

### 14.1 Documentation enclosed

The DTS tank dosing station is supplied along with these instructions.

Separate instructions are provided for the following components if required:

- injection unit
- multi-function valve
- dosing pump (order separately)

### 14.2 Other documentation available

Separate instructions are available for the following components. These are available on the CD supplied or online at [www.Grundfosalldos.com](http://www.Grundfosalldos.com):

- dosing tank/catchment tray
- agitator
- suction line (suction unit)

### 14.3 Assistant for selecting material combinations

The DTS tanks are suitable for the chemicals described below.

Select the material combinations for the components as follows:

| Suitable for:                                                        | Possible combination of dosing head and accessories material*<br>(body/gasket/ball) | Note                                            |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------|
| Sodium hypochlorite (NaClO)                                          | <b>PVC/IV/C</b> , PVC/V/G, PVC/T/T                                                  | Select DDI or DMI dosing pump with Plus3 system |
| Sulphuric acid (H <sub>2</sub> SO <sub>4</sub> ) up to 80 %          | <b>PVC/IV/C</b> , PVC/V/G, PVC/T/T, <b>PP/IV/C</b>                                  | Use PE hoses only                               |
| Hydrogen peroxide (H <sub>2</sub> O <sub>2</sub> )                   | <b>PVC/IV/C</b> , PVC/V/G, PP/V/C                                                   | Select DDI or DMI dosing pump with Plus3 system |
| Antiscalants                                                         | <b>PVC/IV/C</b> , PVC/V/G, PVC/T/T, <b>PP/IV/C</b>                                  |                                                 |
| Biocides                                                             | <b>PVC/IV/C</b> , PVC/V/G, PVC/T/T, <b>PP/IV/C</b>                                  |                                                 |
| Sodium thiosulphate (Na <sub>2</sub> S <sub>2</sub> O <sub>3</sub> ) | <b>PVC/IV/C</b> , PVC/V/G, PVC/T/T, <b>PP/IV/C</b>                                  |                                                 |
| Coagulants and flocculating agents:                                  |                                                                                     |                                                 |
| Iron II/III chloride                                                 | <b>PVC/IV/C</b> , PVC/V/G, PVC/T/T, <b>PP/IV/C</b>                                  | Select DDI or DMI dosing pump with Plus3 system |
| Polyaluminium chloride (PAC)                                         |                                                                                     |                                                 |
| Polyaluminium sulphate                                               |                                                                                     |                                                 |
| Caustic soda (NaOH)                                                  | <b>PP/IE/C</b> , <b>PVC/IE/C</b> , PVC/E/SS, PVC/E/T                                |                                                 |
| Caustic potash (KOH)                                                 | <b>PP/IE/C</b> , <b>PVC/IE/C</b> , PVC/E/SS, PVC/E/T                                |                                                 |
| Potassium permanganate (KMnO <sub>4</sub> )                          | <b>PP/IE/C</b> , <b>PVC/IE/C</b> , PVC/E/T                                          |                                                 |
| Hydrochloric acid (HCl)                                              | <b>PVC/IV/C</b> , PVC/V/G, <b>PP/IE/C</b>                                           |                                                 |
| Phosphoric acid (H <sub>3</sub> PO <sub>4</sub> )                    | <b>PVC/IV/C</b> , PVC/V/G, PVC/T/T, <b>PP/IV/C</b>                                  |                                                 |

\*) the standard combinations are marked in bold

#### Material key for parts in contact with media

|                |                    |
|----------------|--------------------|
| <b>Housing</b> |                    |
| PVC            | Polyvinyl chloride |
| PP             | Polypropylene      |
| <b>Gaskets</b> |                    |
| V              | FKM                |
| E              | EPDM               |
| T              | PTFE               |
| <b>Balls</b>   |                    |
| C              | Ceramic            |
| G              | Glass              |
| T              | PTFE               |
| SS             | Stainless steel    |
| <b>Hoses</b>   |                    |
| PVC            | Polyvinyl chloride |
| PE             | Polyethylene       |
| <b>Tank</b>    |                    |
| PE             | Polyethylene       |



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